

EMMANUEL SAMPSON

ENGINEERING STUDENT • RESEARCHER • CREATIVE
PROFESSIONAL

CONTACT

Williamsburg, Virginia
804.926.0822
ecsampson03@gmail.com
linkedin.com/in/emmanuel-sampson
github.com/emmanuelcsam

EDUCATION

DUAL BACHELOR'S • MAY 2026

William & Mary
Williamsburg, Virginia
BS Physics (Engineering Physics)
BA Philosophy
W&M Hackathon Winner
Catron Scholar, Sharpe Scholar

FILM PRODUCTION CERT • 2024

UCLA School of Theater, Film & Television
Los Angeles, California

KEY SKILLS

Python, C++, MATLAB, Java
Computer Vision & AI/ML

PROFILE

Analytical and innovative engineering physics student with comprehensive experience in optical engineering, computer vision, and advanced instrumentation. Demonstrated expertise in developing AI-powered solutions for fiber-optic systems, conducting cutting-edge research with national laboratories, and leading technical teams. Uniquely combines rigorous scientific training with creative problem-solving abilities, bringing both technical depth and artistic perspective to engineering challenges. Proven track record of winning competitive hackathons and delivering impactful solutions in both academic and industrial settings.

EXPERIENCE

ENGINEER INTERN

MAY 2025—PRESENT

WEINERT Fiber Optics Inc • Richmond, VA

Design and implement AI-powered computer vision systems in Python and C++ for automated fiber-optic defect detection achieving 95% accuracy. Develop real-time medical temperature monitoring sensors for surgical applications. Improved transmission efficiency by 30% for medical laser systems through innovative optical design and signal processing techniques.

PHYSICS RESEARCH ASSISTANT

OCT 2023—PRESENT

Fiber Optics & RF Engineering

CAD & 3D Modeling

Advanced Instrumentation

Data Analysis & Signal Processing

Technical Writing

Film & Photography

INTERESTS

Robotics & Automation

Scientific Research

Creative Writing

Film Production

Photography

Philosophy

Applied Research Center Core Labs • Newport News, VA

Collaborate with CERN, NIST, and CALTECH on low angle reflectivity experiments advancing materials science research. Assembled and operate TRIFT 4 time-of-flight mass spectrometry instrument for high-resolution surface analysis. Develop custom data acquisition systems using Python and LabVIEW. Manage advanced instrumentation including Hitachi S-4700 FE-SEM, Bruker ICON AFM, and Raman spectroscopy systems.

STUDENT ENGINEER

OCT 2022—NOV 2023

W&M Makerspace • Williamsburg, VA

Educated 50+ students on CAD software and advanced fabrication techniques. Managed electro-optic equipment and 3D printing operations for academic and research projects. Led interdisciplinary team projects combining mechanical and electrical engineering solutions.

MANUFACTURING ENGINEER INTERN

MAY 2023—JULY 2023

Build Forward Foundation • Richmond, VA

Maintained and optimized laser cutting and metallurgy equipment for precision manufacturing. Developed motorized vehicle prototypes using CAD and rapid prototyping techniques. Achieved 20% reduction in production time through process optimization and workflow improvements.

CREATIVE PROFESSIONAL

2021—PRESENT

Various Organizations • Virginia

Film Producer at CultureWorks Richmond creating documentary content. Photojournalist for The Virginian-Pilot covering news events. Freelance Writer for Ideapress Publishing and FilmFreeway. Production roles with Two Beagles Studio LLC developing intellectual property. Combine technical expertise with creative storytelling to produce compelling multimedia content.